



● Jianhao Jia

Ph.D. Candidate | Shanghai University of Finance and Economics | Tel: (86)13585838317 |
Email: jianhao.jia@stu.sufe.edu.cn / jianhao.jia@outlook.com | Website: jianhao-jia.github.io

Educational Background

ITCS, Shanghai University of Finance and Economics

2021.9--now

- Advisor: Prof. Nick Gravin.
- Research focus: Algorithmic Game Theory, Online Algorithms, Mechanism Design without Money.
- Thesis topic: Strategic Decision Making.

RIIS, Shanghai University of Finance and Economics

2017.9--2021.6

- RIIS(Research Institute for Interdisciplinary Science). Majoring in interdisciplinary disciplines, involving computer science, finance, statistics etc.

Research Interests

Mechanism Design without Money

- Strategic facility location, strategic clustering, and combinatorial social choice.

Online Algorithms

- Prophet inequalities, online matching, and resource allocation under Markovian demands.

Learning-Augmented Mechanisms

- Mechanism design with predictions, consistency and robustness trade-offs.

Publications

Approximation guarantees of median mechanism in R^d J. Jia, N. Gravin

- Accepted to STOC 2025.
- Resolved a longstanding open problem regarding the tight approximation ratio of the median mechanism in multi-dimensional L_q normed spaces.
- Extended results to the mechanism design with predictions framework.

Online resource allocations in Markov Chains J. Jia, H. Li, K. Liu, Z. Liu, J. Zhou, N. Gravin, Z.G. Tang

- Accepted to WWW 2023.
- Established a new competitive ratio framework for allocation problems where future demand is correlated with the current state.
- Designed algorithms recovering the standard $1/2$ prophet inequality ratio for single-item settings and $1 - o(1)$ ratio for multi-item settings under Markovian demand evolution and mild assumptions.

Working papers

Resource augmented facility location with connections (In Preparation for EC 2026)

Industry Experience (Internship)

Ant Finance, Alibaba Co.

2022.4 - 2023.2

- Collaborated on the project "Online resource allocations in Markov Chains" to model the evolution of financial markets and user behavior.
- Conducted theoretical analysis and applied the Markovian allocation model to large-scale company datasets, demonstrating improved efficiency over standard baselines.
- Resulted in a full paper publication at **WWW 2023**.

Guotai Junan Securities Co.

2019.7 - 2019.8 & 2020.7-2020.8

- Developed Python crawlers to automate data collection from web sources and integrated financial data.
- Performed quantitative analysis on fund products and manager performance using Python.

Service

- TA: Discrete Math (2022, 2024, 2025); Online Decision Making (2024).
- Reviewer: MOR, TCS, AAMAS, SODA; PC member: EC2026.
- Programming skills: python, MATLAB, etc.; Language: English (Proficient- TOEFL 104), Mandarin (Native).